4MA1_2F_ch1_numbers_and_number_system

Nguyen Khac Kiem PhD

1. 4MA1/2FR_Summer_2024_Q25

(a) Write 6.04×10^5 as an ordinary number.

(1)

(b) Write 0.00007 in standard form.

(1)

(c) Work out $\dfrac{7.6 \times 10^{10}}{4 \times 10^5 - 2 \times 10^4}$

Give your answer in standard form.

(2)

2. 4MA1/2FR_Summer_2024_Q24

In a sale, all normal prices are reduced by 17%

The sale price of a fridge is 6225 rupees.

Work out the normal price of the fridge.

.... rupees

(3)

3. 4MA1/2FR_Summer_2024_Q23

$$\frac{3^{-2} \times 3^{5}}{3^{10}} = 3^{n}$$

Find the value of n

(2)

4. 4MA1/2FR_Summer_2024_Q17

Write 1400 as a product of powers of its prime factors.
Show your working clearly.

(3)

5. 4MA1/2FR_Summer_2024_Q16

Footballs are made in a factory.

On Monday, 375 footballs per hour were made in the factory.
On Monday, footballs were made in the factory for 8 hours.

On Tuesday, 20% more footballs were made than on Monday.
On Tuesday, 300 footballs per hour were made in the factory.

Work out for how many hours the factory made footballs on Tuesday.

.... hours

(4)

6. 4MA1/2FR_Summer_2024_Q14

Bilpa, Claudia and David share 1200 euros.

Bilpa gets $\frac{1}{5}$ of the 1200 euros.

Claudia gets 42% of the 1200 euros.

David gets the rest of the 1200 euros.

Given that

the amount of money Bilpa gets : the amount of money David gets = 1 : n

work out the value of n

n = ...

(4)

7. 4MA1/2FR_Summer_2024_Q8

Barsha buys some nails and some bolts.

Each box of nails costs £2.60
Each pack of bolts costs £3.94

Barsha buys 5 boxes of nails and 4 packs of bolts.
He pays with a £50 note.

Work out how much change he should get.

£..

(3)

8. 4MA1/2FR_Summer_2024_Q6

(a) Find the value of 3.4^2

(1)

(b) Find the cube root of 373 248

(1)

(c) Write $7 \times 7 \times 7 \times 7 \times 7 \div 7$ as a single power of 7

(1)

(d) Write one set of brackets in each calculation to make the answer correct.

(i) $5 + 3 \times 2 = 16$

(1)

(ii) $10 - 8 - 10 - 6 \div 2 = 0$

(1)

9. 4MA1/2FR_Summer_2024_Q5

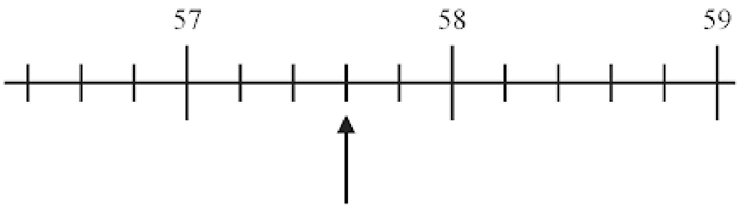
Mina has 5 litres of water in a bottle and some empty cups.
She fills as many cups as possible with water from the bottle.
She fills each cup with 280 millilitres of water.

Work out how many cups Mina completely fills.

(3)

10. 4MA1/2FR_Summer_2024_Q3

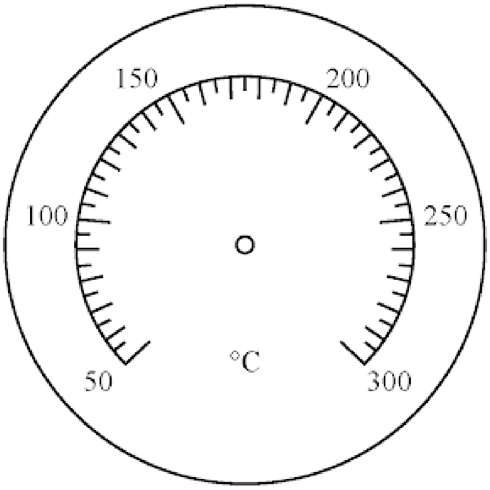
The diagram shows part of a number line.



(a) Write down the number marked with the arrow.

(1)

(b) On the diagram below, draw an arrow to show a temperature of 240°C



(1)

(c) Write the number 0.786 correct to 2 decimal places.

(1)

11. 4MA1/2FR_Summer_2024_Q1

(a) Write these numbers in order of size.
Start with the smallest number.

122 5 150 71 39

(1)

(b) Write these decimals in order of size.
Start with the smallest decimal.

0.7 0.074 3.77 0.37 0.13

(1)

(c) Write in figures the number five thousand and eighty four.

(1)

(d) Write down the value of the 3 in the number 1324

(1)

12. 4MA1/2F_Summer_2024_Q25

Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of 4 years.
Give your answer correct to the nearest koruna.

.... koruna

(3)

13. 4MA1/2F_Summer_2024_Q24

Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

(5)

14. 4MA1/2F_Summer_2024_Q20

Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

(3)

15. 4MA1/2F_Summer_2024_Q16

Use your calculator to work out the value of

$$\frac{17.9}{8.61 + 2.36} - 1.2^2$$

Give your answer as a decimal.
Write down all the figures on your calculator display.

(2)

16. 4MA1/2F_Summer_2024_Q12

Shane buys a clock in Australia.

The clock costs 232 Australian dollars.
In India, an identical clock costs 12 420 Indian rupees.

The exchange rate is 1 Australian dollar = 54 Indian rupees

The clock costs more in Australia than in India.

Work out how much more.
You must give the units of your answer.

(3)

17. 4MA1/2F_Summer_2024_Q8

Bargain Crafts and Art's Store each have a special offer on tins of crayons.

Bargain Crafts	Art's Store
Crayons \$4.20 per tin	Pack of 5 tins of crayons for \$18
Special offer	**Special offer**
Pay for 2 tins get 1 tin free	25% off each pack of 5 tins

Heidi buys 30 tins of crayons from Bargain Crafts using the special offer.
Amir buys 30 tins of crayons from Art's Store using the special offer.

Work out the difference between the amount that Heidi pays and the amount that Amir pays.

\$.......................................

(4)

18. 4MA1/2F_Summer_2024_Q6

There are 150 animals on a farm.

Of these animals

19 are sheep
32 are goats
3 are dogs

The rest of the animals are chickens.

Write the number of chickens as a fraction of the total number of animals.
Give your fraction in its simplest form.

(3)

19. 4MA1/2F_Summer_2024_Q3

(a) Write these numbers in numerical order.
Start with the smallest number.

–77 39 –89 43 –6

(1)

(b) Write these decimals in order of size.
Start with the smallest decimal.

0.134 0.12 0.145 0.017 0.3

(1)

(c) Write 0.7 as a percentage.

....%

(1)

(d) Write $\frac{27}{100}$ as a decimal.

(1)

There are 60 plants on a plant stall.

$\frac{7}{10}$ of the plants are vegetable plants.

(e) Work out how many of the plants are **not** vegetable plants.

(2)

20. 4MA1/2FR_Winter_2023_Q23

Teresa invests \$2000 for 3 years in a savings account.
She gets 4% each year compound interest.

(a) How much money will Teresa have in her savings account at the end of 3 years?
Give your answer correct to the nearest dollar.

\$..
(3)

Sam invested \$$T$
The value of his investment decreased by 9% each year.

At the end of the first year, the value of Sam's investment was \$1365

(b) Work out the value of T

....
(3)

21. 4MA1/2FR_Winter_2023_Q22

60 students sat a Mathematics exam.

The mean mark for the 32 students in Class A was 55
The mean mark for the 28 students in Class B was 52

Find the mean mark for all 60 students.

....

3 marks

22. 4MA1/2FR_Winter_2023_Q21

(a) Find the highest common factor (HCF) of 200 and 420

....

(2)

$A = 2^3 \times 3 \times 5 \times 7^2$
$B = 2 \times 3^2 \times 7$
$C = 3 \times 5^2 \times 11$

(b) Find the lowest common multiple (LCM) of A, B and C
Write your answer as a product of powers of prime factors.

....

(2)

23. 4MA1/2FR_Winter_2023_Q17

$-8 < 2y \leqslant 2$

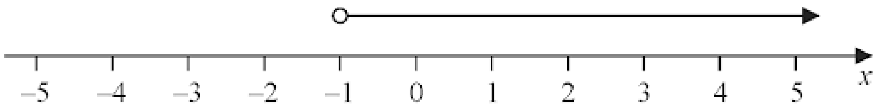
y is an integer.

(a) Find all the possible values of y

....

(2)

(b) Write down the inequality shown on the number line.



....

(1)

24. 4MA1/2FR_Winter_2023_Q15

Show that $4\frac{2}{3} \div 1\frac{5}{6} = 2\frac{6}{11}$

3 marks

25. 4MA1/2FR_Winter_2023_Q11

Asif has 200 beads.

Asif gives $\frac{1}{4}$ of the 200 beads to Bernadette.

Asif gives $\frac{2}{5}$ of the 200 beads to Claudio.

Asif gives 43 beads to Derek.

What fraction of the 200 beads does Asif have left?

....

4 marks

26. 4MA1/2FR_Winter_2023_Q10

Hermann changed £500 into euros.
The exchange rate was £1 = 1.18 euros.

(a) Work out how much money, in euros, Hermann received.

.... euros

(2)

Anita changed \$350 into pounds (£)
The exchange rate was £1 = \$1.40

(b) Work out how much money, in pounds (£), Anita received.

£..

(2)

27. 4MA1/2FR_Winter_2023_Q9

(a) Write the time 8 30 pm using the 24-hour clock.

....
(1)

Ella started watching television at 10 50 am.

Ella watched

a comedy programme lasting 45 minutes
a sports programme lasting 1 hour 10 minutes
a history programme

There were no breaks and no advertisements between the programmes.
Ella finished watching television at 2 20 pm.

(b) How long did the history programme last?
Give your answer in minutes.

.... minutes
(3)

28. 4MA1/2FR_Winter_2023_Q8

Linford is going to take part in an athletics competition.
He can choose one event from List **A** and one event from List **B**

List A	**List B**
Discus (D) Hammer (H) Javelin (J) Shot Put (S)	Long Jump (L) Pole Vault (P) Relay (R)

Write down all the possible combinations Linford can choose.

2 marks

29. 4MA1/2FR_Winter_2023_Q7

In a field, there are 60 sheep and 24 cows.

(a) Find the ratio of the number of sheep to the number of cows.
Give your ratio in its simplest form.

....

(2)

In a barn, there are only white ducks and brown ducks.
In the barn, the ratio

number of white ducks : number of brown ducks = 3 : 7

(b) What fraction of the ducks in the barn are white?

....

(1)

Giles and Sarah share some bales of hay in the ratio 11 : 4

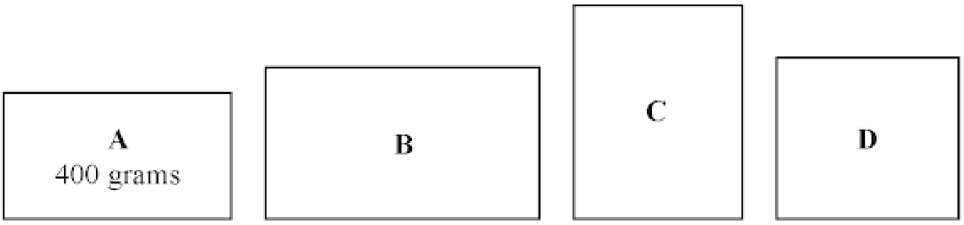
Sarah receives 20 bales of hay.

(c) Work out how many bales of hay are shared in total.

....

(3)

Pat has 4 parcels **A**, **B**, **C** and **D**



The weight of parcel **A** is 400 grams.

The weight of parcel **B** is 350 grams more than the weight of parcel **A**
The weight of parcel **C** is twice the weight of parcel **A**

The total weight of the 4 parcels is 2.5 kilograms.

Work out the weight, in grams, of parcel **D**

.... grams

4 marks

31. 4MA1/2FR_Winter_2023_Q1

(a) Write these numbers in order of size.
Start with the smallest number.

171 490 84 105 233

....

(1)

(b) Write in figures the number five thousand, one hundred and two.

....

(1)

(c) Write down the value of the 3 in the number 7439

....

(1)

(d) Write the number 651 correct to the nearest hundred.

....

(1)

Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

Bank **G**	Bank **H**
1.6% per year compound interest	2.9% interest added after two years

The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

How much more?

.... Swiss francs

4 marks

The weight of a cake is 2.75 kg, correct to 2 decimal places.

(a) Write down the lower bound of the weight of the cake.

.... kg
(1)

(b) Write down the upper bound of the weight of the cake.

.... kg
(1)

Penny has worked out $\dfrac{81.3 \times 59.2}{1.9^2}$ on her calculator.

Her answer is 13 332.299 17

Penny's answer is not sensible.

(c) By rounding each number to one significant figure, work out a suitable estimate to show that her answer is not sensible.
Show your working clearly.

(2)

34. 4MA1/2F_Winter_2023_Q17

Change a speed of 90 kilometres per hour to a speed in metres per second.
Show your working clearly.

.... m/s

3 marks

35. 4MA1/2F_Winter_2023_Q16

Show that $3\frac{5}{7} \div 1\frac{5}{8} = 2\frac{2}{7}$

3 marks

There are 380 students in a Sixth Form.
The students are either in the Upper Sixth or in the Lower Sixth.

The number of students in the Upper Sixth is 20 fewer than the number of students in the Lower Sixth.

$\frac{2}{5}$ of the Upper Sixth students study mathematics.

32% of the Lower Sixth students study mathematics.

Work out the total number of students in the Sixth Form who study mathematics.

....

4 marks

37. 4MA1/2F_Winter_2023_Q12

Last season, Alisha and Jaya scored goals for their team in the ratio 4 : 7
Jaya scored 39 more goals than Alisha.

Work out the number of goals Alisha scored.

....

3 marks

(a) Work out the value of $(4 + 3 + 6)^2$

....
(1)

$64 = 4^n$

(b) Write down the value of n

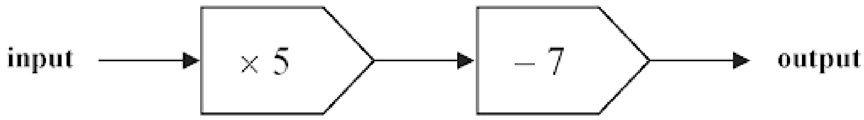
$n =$ ....
(1)

(c) Work out the value of $\dfrac{\sqrt{9.3 + 2.8^3}}{3.2 \times 1.2}$

Write down all the figures on your calculator display.

....
(2)

Here is a number machine.

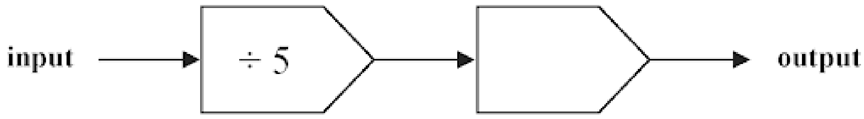


(a) Work out the output when the input is 9

....

(1)

Here is a different number machine.



When the input is 30 the output is 18

(b) Find a suitable way to complete the number machine.

(1)

The following rule is used to work out the total cost, in euros, of hiring a cement mixer.

Total cost = 8 euros per day plus 5 euros

James hires a cement mixer for 3 days.

(c) Work out the cost to James of hiring the cement mixer.

.... euros

(1)

The cost to Sophia of hiring a cement mixer is 61 euros.

(d) For how many days does Sophia hire the cement mixer?

.... days

(2)

(a) Write 5 15 pm using the 24-hour clock.

....
(1)

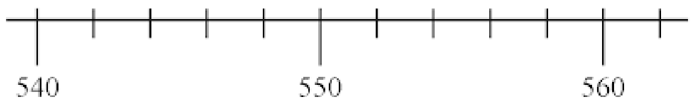
Roberta goes out for a walk.
She leaves home at 16 35
She arrives back home at 20 15 on the same day.

(b) Work out for how much time Roberta is out for her walk.

.... hours .... minutes
(2)

41. 4MA1/2F_Winter_2023_Q3

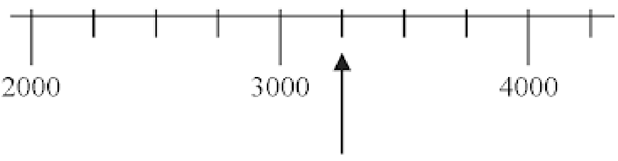
Here is a number scale.



(a) On the scale, mark with an arrow (↑) the number 554

(1)

Here is a different number scale.

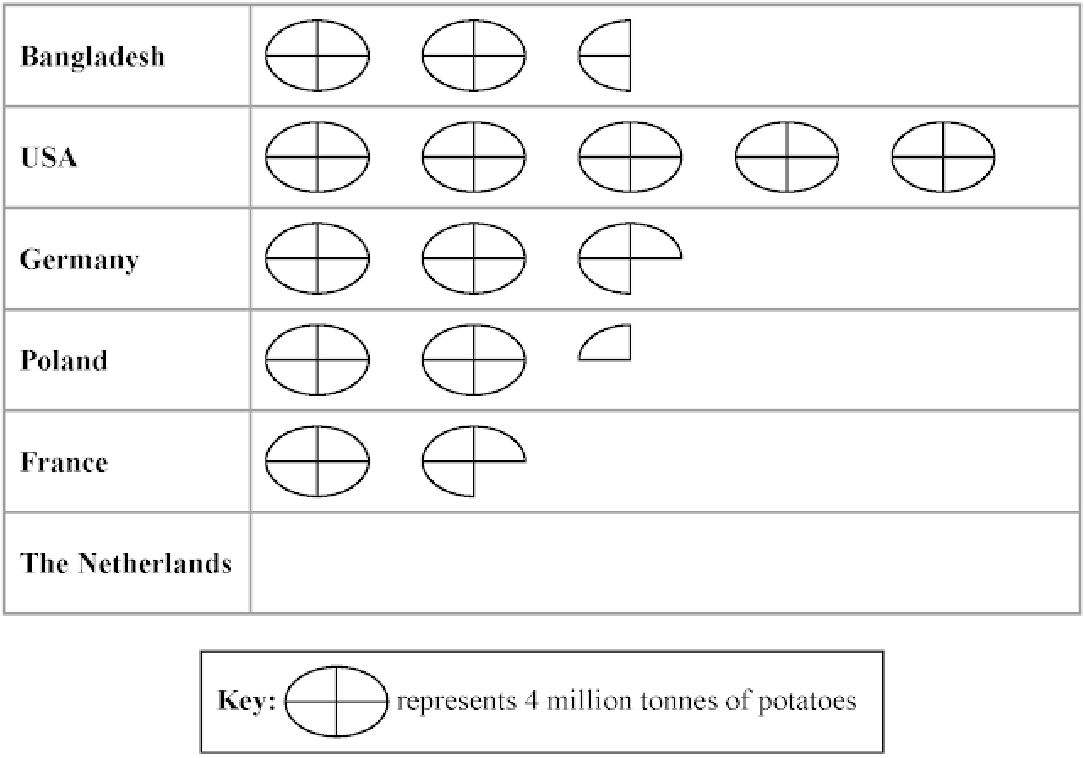


(b) Write down the number shown marked by the arrow.

....

(1)

The pictogram shows information about the total weight of potatoes grown last year in each of five countries.



The pictogram shows one country where the total weight of potatoes grown last year was 20 million tonnes.

(a) Which country?

....

(1)

Last year, the weight of potatoes grown in The Netherlands was 6 million tonnes.

(b) Show this information on the pictogram.

(1)

(c) Work out the total weight of potatoes grown in Germany **and** in France last year.

.... million tonnes

(2)

Here are four cards.
Each card has a number on it.

5	7	6	3

These four cards are arranged to make the number 5763

(a) Arrange the four cards to make the smallest possible number.

(1)

(b) Arrange the four cards to make the largest possible **even** number.

(1)

(c) Arrange two of the cards to make a prime number.

(1)

(d) Arrange two of the cards to make a multiple of 8

(1)

Gemara works as a taxi driver.
Last week, he recorded the following information about the distances he drove.

For the 5 days from Monday to Friday, the mean number of kilometres he drove was 104
For the 7 days from Monday to Sunday, the mean number of kilometres he drove was 127

On Saturday, Gemara drove 132 kilometres.

Work out the number of kilometres he drove on Sunday.

.... kilometres

3 marks

In his previous job, Pierre was paid 400 euros in total for working a 5-day week.

In his new job, Pierre is paid 14 euros per hour.
In his new job, Pierre works for 7 hours each day for a 5-day week.

(a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week.

....%
(4)

Marie changes her job.

Her salary decreases by 6%
Her new salary is 23 030 euros.

(b) Work out Marie's salary before she changes her job.

.... euros
(3)

46. 4MA1/2FR_Summer_2023_Q18

Anjali travels on the Eurostar train from Paris to Amsterdam.

The distance the train travels between Paris and Amsterdam is 515 km.
The time taken by the train to travel between Paris and Amsterdam is 3 hours 18 minutes.

Work out the average speed of the train.
Give your answer in km/h correct to the nearest whole number.

.... km/h

3 marks

Sandeep wants to buy some packets of pens and some boxes of pencils for his stationery shop.

Each packet of pens contains 9 pens.
Each box of pencils contains 12 pencils.

Each packet of pens costs £7.60
Each box of pencils costs £4.80

Sandeep can only buy full packets of pens and full boxes of pencils.
He wants to buy exactly the same number of pens as pencils.

Work out the minimum amount Sandeep needs to pay.

£..

4 marks

One month, Abbad raised money for charity by being sponsored to run and to walk.
His target was to raise 700 dirhams.

During the month, Abbad recorded a total distance of 135 km by running and by walking.
The ratio

number of km he ran : number of km he walked = 2 : 7

Abbad received 8 dirhams for each km he ran and 5 dirhams for each km he walked.

Abbad raised more money than his 700 dirhams target.

How much more?

.... dirhams

4 marks

49. 4MA1/2FR_Summer_2023_Q13

One week, an online company carried out a total of 175 deals.
49 of these deals were in yen.

Work out the percentage of deals carried out in yen.

....%

2 marks

50. 4MA1/2FR_Summer_2023_Q11

180 passengers travelled either to Seoul or to Tokyo.
They travelled either business class or economy class.

Of these 180 passengers

51 travelled business class to Seoul
69 travelled economy class to Tokyo
86 in total travelled business class

Use this information to complete the two-way table.

	Seoul	**Tokyo**	**Total**
Business class			
Economy class			
Total			

3 marks

51. 4MA1/2FR_Summer_2023_Q10

Haneul buys 120 packs of cherries.
Each pack of cherries that Haneul buys costs 28 Malaysian ringgits.

Haneul sells $\frac{4}{5}$ of the 120 packs of cherries.

He sells each of these packs for 46 Malaysian ringgits.

The remaining packs are unsold.

Work out Haneul's total profit.

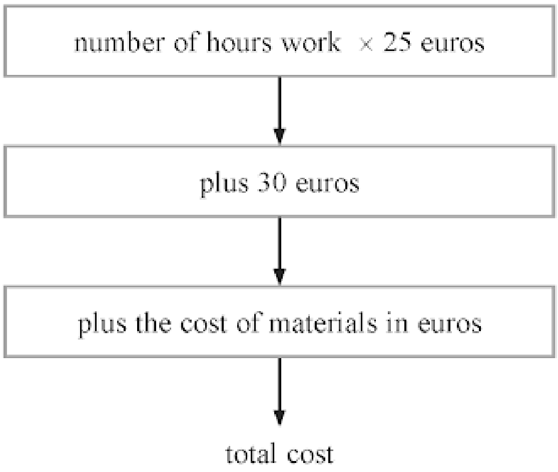
.... Malaysian ringgits

4 marks

52. 4MA1/2FR_Summer_2023_Q8

Georgia is a plumber.

She uses this rule to work out the total cost, in euros, of her customers' bills.



Georgia does some plumbing work for Tallulah.

She works for 3 hours.
The cost of the materials she uses is 42 euros.

(a) Work out the total cost of Tallulah's bill.

.... euros

(2)

Georgia also does some plumbing work for Sam.

The cost of the materials she uses is 65 euros.
The total cost of Sam's bill is 220 euros.

(b) Work out the number of hours that Georgia works for Sam.

.... hours

(3)

53. 4MA1/2FR_Summer_2023_Q7

Gerhard is travelling from Zurich to Munich.

The table shows the exchange rate in each city.

Zurich	**Munich**
1 Swiss franc = 0.90 euros	1 euro = 1.08 Swiss francs

Gerhard wants to change 1620 Swiss francs to euros.

He would receive more euros if he changes his Swiss francs in Munich than he would receive if he changes his Swiss francs in Zurich.

How many more?

.... euros

4 marks

54. 4MA1/2FR_Summer_2023_Q5

Here is a list of fractions.

$$\frac{4}{5} \qquad \frac{3}{8} \qquad \frac{2}{3} \qquad \frac{12}{15} \qquad \frac{7}{10}$$

One fraction from the list, when written as a decimal, is **not** a terminating decimal.

(a) Write down this fraction.

....

(1)

Two fractions from the list are equivalent.

(b) Write down these two fractions.

.... and ....

(1)

(c) Write $\frac{7}{10}$ as a decimal.

....

(1)

The table shows the mean daytime temperature for each of five planets.

Planet	**Temperature (°C)**
Jupiter	–160
Mars	–30
Mercury	430
Neptune	–210
Venus	470

(a) Which of these five planets has the lowest mean daytime temperature?

....
(1)

(b) Work out the difference between the mean daytime temperature for Jupiter and the mean daytime temperature for Venus.

.... °C
(1)

20	21	23	25	27	29

From the numbers in the box, write down

(a) (i) the even number,

....
(1)

(ii) the square number.

....
(1)

prime	square	cube	factor	product

(b) Complete the following statements by writing a suitable word from the box on each of the dotted lines.

(i) 343 is the .... of 7
(1)

(ii) 9 is a .... of 63
(1)

57. 4MA1/2F_Summer_2023_Q26

Team **A** and Team **B** take part in a quiz league.

After 11 rounds, Team **A** has a mean score per round of 17
After 9 rounds, Team **B** has a mean score per round of 18

Both teams take part in a further round.
After this round, both teams have a mean score per round of 18.5

In the further round, Team **A** scored more points than Team **B**.

How many more?

....

4 marks

58. 4MA1/2F_Summer_2023_Q25

Change a speed of 27 kilometres per hour to a speed in metres per second.

.... m / s

3 marks

59. 4MA1/2F_Summer_2023_Q24

Charlotte buys a painting for \$680
The value of the painting increases by 4% each year.

Work out the value of the painting at the end of 3 years.
Give your answer correct to the nearest \$

\$..

3 marks

60. 4MA1/2F_Summer_2023_Q22

(a) Write 76 000 000 in standard form.

....
(1)

(b) Write 5.4×10^{-4} as an ordinary number.

....
(1)

Nancy has some coins with a total value of 85 pence.
She has only 2 pence coins and 5 pence coins.
The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1 : 3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

....

4 marks

62. 4MA1/2F_Summer_2023_Q17

Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

3 marks

63. 4MA1/2F_Summer_2023_Q15

Finn is asked to find the value of $5 + 3^2 + 12$

Here is his working and his answer.

$$\begin{aligned} 5 + 3^2 + 12 &= 8^2 + 12 \\ &= 64 + 12 \\ &= 76 \end{aligned}$$

Finn's answer is wrong.

(a) Explain what Finn has done wrong in his working.

....

....

(1)

(b) Write one pair of brackets in this calculation so that the answer is correct.

$$2 \times 6 - 4^2 - 14 = 10$$

(1)

(c) Work out the value of $x^2 + 5y$ when $x = -3$ and $y = 2$

....

(2)

64. 4MA1/2F_Summer_2023_Q14

1 dollar = 0.85 euros
1 dollar = 6.45 yuan

(a) Change 720 dollars to euros.

.... euros
(1)

(b) Change 2580 yuan to euros.

.... euros
(2)

65. 4MA1/2F_Summer_2023_Q10

Adriana makes some cupcakes.

She sells each cupcake for £2.20

Mike buys 6 of these cupcakes.
He pays with a £20 note.

(a) Work out how much change Mike should receive.

£..
(2)

Greta sells balloons.

She sells each balloon for £0.85

Stefan has £50 to spend on balloons.

(b) Work out the greatest number of balloons he can buy.

....
(2)

A hall has 26 rows of seats.
There are 14 seats in each row.

Annie sells tickets for $\frac{3}{4}$ of the seats in the hall for a concert.
She sells each ticket for 15 euros.

Work out the total amount Annie gets from selling tickets.

.... euros

3 marks

67. 4MA1/2F_Summer_2023_Q2

Here is a list of numbers.

9 11 13 15 21 60

From the numbers in the list, write down

(a) the even number

....
(1)

(b) the square number

....
(1)

(c) the multiple of 7

....
(1)

(d) the factor of 30

....
(1)

68. 4MA1/2F_Summer_2023_Q1

(a) Write in figures the number seven thousand and fifty four.

....

(1)

(b) Write the number 78 263 correct to the nearest 100

....

(1)

(c) Write down the value of the 7 in the number 673 000

....

(1)

(d) Write a number in the box to make the calculation correct.

$$9 \times \boxed{} = 18\,000$$

(1)

(e) Write a number in the box to make the calculation correct.

$$53.7 \div \boxed{} = 0.005\,37$$

(1)

69. 4MA1/2FR_Winter_2022_Q25

A scientist is investigating the weight of 50 tigers.

Here is some information about these tigers.

	Type of tiger	
	Siberian	**Bengal**
Number of tigers	22	28
Mean weight of tigers (kg)	260	

The mean weight of all 50 tigers is 218 kg

Work out the mean weight of the Bengal tigers.

.... kg

3 marks

70. 4MA1/2FR_Winter_2022_Q23

(a) Write down the value of x^0

....
(1)

Given that $2^{-3} \times 2^{9} = 2^{n}$

(b) find the value of n

$n =$ ....
(1)

Given that $\dfrac{7^{206} \times 7^{m}}{7^{214}} = 7^{-3}$

(c) find the value of m

$m =$ ....
(2)

71. 4MA1/2FR_Winter_2022_Q22

Shane invests 7200 dollars for 3 years in a savings account.
He gets 2.5% per year compound interest.

How much money will Shane have in his savings account at the end of 3 years?
Give your answer to the nearest dollar.

.... dollars

3 marks

Mary saves for a holiday each year.

In 2020 she saved a total of \$720
In 2021, each month she saved \$78

The total amount Mary saved in 2021 was P% more than the total she saved in 2020

(a) Work out the value of P

....

(4)

73. 4MA1/2FR_Winter_2022_Q16

(a) Show that $\frac{3}{8} \div \frac{27}{32} = \frac{4}{9}$

(2)

(b) Show that $\frac{5}{6} - \frac{3}{8} = \frac{11}{24}$

(2)

74. 4MA1/2FR_Winter_2022_Q12

Maria is going to make some flapjacks.

Here are four of the ingredients that she will use.

175 g butter
175 g syrup
175 g sugar
330 g oats

What percentage of these four ingredients is oats?
Give your answer correct to 3 significant figures.

.... %

3 marks

75. 4MA1/2FR_Winter_2022_Q11

At school each week, Gabriella has to play a sport on Monday and a sport on Thursday.

The table shows the sports from which she can choose on Monday and the sports from which she can choose on Thursday.

Monday	**Thursday**
Tennis (T)	Volleyball (V)
Netball (N)	Badminton (B)
Hockey (H)	Tennis (T)

Write down all the possible combinations of these sports that Gabriella can play in one week.

....

....

....

2 marks

76. 4MA1/2FR_Winter_2022_Q10

77. 4MA1/2FR_Winter_2022_Q8

78. 4MA1/2FR_Winter_2022_Q7

79. 4MA1/2FR_Winter_2022_Q5

80. 4MA1/2FR_Winter_2022_Q1

81. 4MA1/2F_Winter_2022_Q23

82. 4MA1/2F_Winter_2022_Q22

83. 4MA1/2F_Winter_2022_Q17

84. 4MA1/2F_Winter_2022_Q15

85. 4MA1/2F_Winter_2022_Q14

86. 4MA1/2F_Winter_2022_Q13

87. 4MA1/2F_Winter_2022_Q10

88. 4MA1/2F_Winter_2022_Q8

89. 4MA1/2F_Winter_2022_Q7

90. 4MA1/2F_Winter_2022_Q4

91. 4MA1/2F_Winter_2022_Q2

92. 4MA1/2F_Winter_2022_Q1

93. 4MA1/2FR_Summer_2022_Q26

94. 4MA1/2FR_Summer_2022_Q25

95. 4MA1/2FR_Summer_2022_Q19

96. 4MA1/2FR_Summer_2022_Q15

97. 4MA1/2FR_Summer_2022_Q10

98. 4MA1/2FR_Summer_2022_Q8

99. 4MA1/2FR_Summer_2022_Q6

100. 4MA1/2FR_Summer_2022_Q5

101. 4MA1/2FR_Summer_2022_Q3

102. 4MA1/2FR_Summer_2022_Q2

103. 4MA1/2FR_Summer_2022_Q1

104. 4MA1/2F_Summer_2022_Q28

105. 4MA1/2F_Summer_2022_Q25

106. 4MA1/2F_Summer_2022_Q24

107. 4MA1/2F_Summer_2022_Q23

108. 4MA1/2F_Summer_2022_Q19

109. 4MA1/2F_Summer_2022_Q15

110. 4MA1/2F_Summer_2022_Q13

111. 4MA1/2F_Summer_2022_Q12

112. 4MA1/2F_Summer_2022_Q9

113. 4MA1/2F_Summer_2022_Q8

114. 4MA1/2F_Summer_2022_Q6

115. 4MA1/2F_Summer_2022_Q1

116. 4MA1/2FR_Winter_2021_Q19

117. 4MA1/2FR_Winter_2021_Q17

118. 4MA1/2FR_Winter_2021_Q10

119. 4MA1/2FR_Winter_2021_Q8

120. 4MA1/2FR_Winter_2021_Q7

121. 4MA1/2FR_Winter_2021_Q6

122. 4MA1/2FR_Winter_2021_Q5

123. 4MA1/2FR_Winter_2021_Q3

124. 4MA1/2FR_Winter_2021_Q1

125. 4MA1/2F_Winter_2021_Q24

126. 4MA1/2F_Winter_2021_Q23

127. 4MA1/2F_Winter_2021_Q20

128. 4MA1/2F_Winter_2021_Q17

129. 4MA1/2F_Winter_2021_Q15

130. 4MA1/2F_Winter_2021_Q12

131. 4MA1/2F_Winter_2021_Q8

132. 4MA1/2F_Winter_2021_Q4

133. 4MA1/2F_Winter_2021_Q3

134. 4MA1/2F_Winter_2021_Q1

135. 4MA1/2F_Summer_2021_Q25

136. 4MA1/2F_Summer_2021_Q20

137. 4MA1/2F_Summer_2021_Q16

138. 4MA1/2F_Summer_2021_Q15

139. 4MA1/2F_Summer_2021_Q14

140. 4MA1/2F_Summer_2021_Q12

141. 4MA1/2F_Summer_2021_Q7

142. 4MA1/2F_Summer_2021_Q6

143. 4MA1/2F_Summer_2021_Q5

144. 4MA1/2F_Summer_2021_Q1

145. 4MA1/2FR_Winter_2020_Q24

146. 4MA1/2FR_Winter_2020_Q22

147. 4MA1/2FR_Winter_2020_Q20

148. 4MA1/2FR_Winter_2020_Q17

149. 4MA1/2FR_Winter_2020_Q14

150. 4MA1/2FR_Winter_2020_Q13

151. 4MA1/2FR_Winter_2020_Q11

152. 4MA1/2FR_Winter_2020_Q6

153. 4MA1/2FR_Winter_2020_Q4

154. 4MA1/2FR_Winter_2020_Q1

155. 4MA1/2F_Winter_2020_Q26

156. 4MA1/2F_Winter_2020_Q25

157. 4MA1/2F_Winter_2020_Q24

158. 4MA1/2F_Winter_2020_Q19

159. 4MA1/2F_Winter_2020_Q18

160. 4MA1/2F_Winter_2020_Q12

161. 4MA1/2F_Winter_2020_Q11

162. 4MA1/2F_Winter_2020_Q10

163. 4MA1/2F_Winter_2020_Q2

164. 4MA1/2F_Winter_2020_Q1

165. 4MA1/2FR_Summer_2020_Q21

166. 4MA1/2FR_Summer_2020_Q19

167. 4MA1/2FR_Summer_2020_Q18

168. 4MA1/2FR_Summer_2020_Q16

169. 4MA1/2FR_Summer_2020_Q15

170. 4MA1/2FR_Summer_2020_Q11

171. 4MA1/2FR_Summer_2020_Q10

172. 4MA1/2FR_Summer_2020_Q9

173. 4MA1/2FR_Summer_2020_Q8

174. 4MA1/2FR_Summer_2020_Q6

175. 4MA1/2FR_Summer_2020_Q1

176. 4MA1/2F_Summer_2020_Q20

177. 4MA1/2F_Summer_2020_Q19

178. 4MA1/2F_Summer_2020_Q17

179. 4MA1/2F_Summer_2020_Q15

180. 4MA1/2F_Summer_2020_Q12

181. 4MA1/2F_Summer_2020_Q11

182. 4MA1/2F_Summer_2020_Q10

183. 4MA1/2F_Summer_2020_Q9

184. 4MA1/2F_Summer_2020_Q8

185. 4MA1/2F_Summer_2020_Q5

186. 4MA1/2F_Summer_2020_Q4

187. 4MA1/2F_Summer_2020_Q2

188. 4MA1/2F_Summer_2020_Q1

189. 4MA1/2FR_Winter_2019_Q24

190. 4MA1/2FR_Winter_2019_Q21

191. 4MA1/2FR_Winter_2019_Q20

192. 4MA1/2FR_Winter_2019_Q18

193. 4MA1/2FR_Winter_2019_Q12

194. 4MA1/2FR_Winter_2019_Q9

195. 4MA1/2FR_Winter_2019_Q8

196. 4MA1/2FR_Winter_2019_Q5

197. 4MA1/2FR_Winter_2019_Q3

198. 4MA1/2FR_Winter_2019_Q2

199. 4MA1/2FR_Winter_2019_Q1

200. 4MA1/2F_Winter_2019_Q22

201. 4MA1/2F_Winter_2019_Q21

202. 4MA1/2F_Winter_2019_Q20

203. 4MA1/2F_Winter_2019_Q17

204. 4MA1/2F_Winter_2019_Q14

205. 4MA1/2F_Winter_2019_Q8

206. 4MA1/2F_Winter_2019_Q3

207. 4MA1/2F_Winter_2019_Q1

208. 4MA1/2FR_Summer_2019_Q25

209. 4MA1/2FR_Summer_2019_Q24

210. 4MA1/2FR_Summer_2019_Q23

211. 4MA1/2FR_Summer_2019_Q22

212. 4MA1/2FR_Summer_2019_Q18

213. 4MA1/2FR_Summer_2019_Q14

214. 4MA1/2FR_Summer_2019_Q9

215. 4MA1/2FR_Summer_2019_Q8

216. 4MA1/2FR_Summer_2019_Q6

217. 4MA1/2FR_Summer_2019_Q5

218. 4MA1/2FR_Summer_2019_Q1

219. 4MA1/2F_Summer_2019_Q19

220. 4MA1/2F_Summer_2019_Q18

221. 4MA1/2F_Summer_2019_Q16

222. 4MA1/2F_Summer_2019_Q11

223. 4MA1/2F_Summer_2019_Q8

224. 4MA1/2F_Summer_2019_Q5

225. 4MA1/2F_Summer_2019_Q1

226. 4MA1/2FR_Summer_2018_Q22

227. 4MA1/2FR_Summer_2018_Q20

228. 4MA1/2FR_Summer_2018_Q18

229. 4MA1/2FR_Summer_2018_Q17

230. 4MA1/2FR_Summer_2018_Q13

231. 4MA1/2FR_Summer_2018_Q11

232. 4MA1/2FR_Summer_2018_Q10

233. 4MA1/2FR_Summer_2018_Q8

234. 4MA1/2FR_Summer_2018_Q6

235. 4MA1/2FR_Summer_2018_Q4

236. 4MA1/2FR_Summer_2018_Q2

237. 4MA1/2FR_Summer_2018_Q1

238. 4MA1/2F_Summer_2018_Q20

239. 4MA1/2F_Summer_2018_Q17

240. 4MA1/2F_Summer_2018_Q16

241. 4MA1/2F_Summer_2018_Q13

242. 4MA1/2F_Summer_2018_Q11

243. 4MA1/2F_Summer_2018_Q7

244. 4MA1/2F_Summer_2018_Q5

245. 4MA1/2F_Summer_2018_Q3

246. 4MA1/2F_Summer_2018_Q1